A UNIVERSALLY CLOSED-ON-POINTS MAP OVER THE BASE DETECTS RELATIVE PSEUDO-COHERENCE THROUGH ITS PUSHFORWARD

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 16948-16955 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 16957-16992, the complete author proof (34 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The author writes rad for the Jacobson radical in this proof; retained verbatim.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Affineness of point-closed maps locally of finite type (Morphisms, lemma-homeomorphism-affine), the affine-space test for relative pseudo-coherence (same chapter), and invariance under closed pushforwards (Derived Categories of Schemes, lemma-closed-push-pseudo-coherent).

Source reference. Original lemma 09UJ: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16837–16847. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04DE: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16992–16992. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 09VA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 16992–16992. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $i : X \to Y$ morphism of schemes locally of finite type over a base scheme S. Assume that i induces a homeomorphism of X with a closed subset of Y. Let E be an object of $D(\mathcal{O}_X)$. Then E is m-pseudo-coherent relative to S if and only if Ri_*E is m-pseudo-coherent relative to S.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. By Morphisms, Lemma 04DE the morphism i is affine. Thus we may assume S, Y, and X are affine. Say $S = \mathrm{Spec}(R)$, $Y = \mathrm{Spec}(A)$, and $X = \mathrm{Spec}(B)$. The condition means that $A/\mathrm{rad}(A) \to B/\mathrm{rad}(B)$ is surjective; here $\mathrm{rad}(A)$ and $\mathrm{rad}(B)$ denote the Jacobson radical of A and B. As B is of finite type over A, we can find $b_1, \ldots, b_m \in \mathrm{rad}(B)$ which generate B as an A-algebra. Say $b_j^N = 0$ for all j. Consider the diagram of rings

$$\begin{array}{ccccc} B & \longleftarrow & R[x_i, y_j]/(y_j^N) & \longleftarrow & R[x_i, y_j] \\ \uparrow & & \uparrow & \nearrow & \\ A & \longleftarrow & R[x_i] & & \end{array}$$

which translates into a diagram

$$\begin{array}{ccccc} X & \longrightarrow & T & \longrightarrow & \mathbf{A}_S^{n+m} \\ \downarrow & & \downarrow & \swarrow & \\ Y & \longrightarrow & \mathbf{A}_S^n & & \end{array}$$

of affine schemes. By Lemma 09UJ we see that E is m-pseudo-coherent relative to S if and only if its pushforward to $\mathbf{A}_S^{n+m}$ is m-pseudo-coherent. By Derived Categories of Schemes, Lemma 09VA we see that this is true if and only if its pushforward to T is m-pseudo-coherent. The same lemma shows that this holds if and only if the pushforward to $\mathbf{A}_S^n$ is m-pseudo-coherent. Again by Lemma 09UJ this holds if and only if Ri_*E is m-pseudo-coherent relative to S. $\square$